Yaqin Si

Raleigh, NC

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Research Summary & Core Skill Set

- **Research Summary** Ph.D. candidate in Bioinformatics with expertise in statistical genetics and large-scale analysis of EHR and claims data. Developed CNVreg, a novel method for CNV-profile regression, along with scalable pipelines for joint CNV/SNP analysis. Released a CRAN R package for CNV association analysis and applied these methods to identify CNVs associated with Alzheimer's disease. Research interests include large-scale data analysis, machine learning, and causal inference.

- Excellent Programming: Data cleaning, visualization, algorithm implementation, reproducible pipelines, and package development.
- Tools: R, Python, SAS, Julia, SQL, TreeAge, Tableau, GitHub, Shell scripts; Linux; HPC platform; AWS
- Statistics & Machine Learning: Advanced statistical modeling, simulation studies, experimental design, deep learning and large language models
- Data Expertise: genomics data (WGS, CNV/SV, SNP) and real-world EHR & Claims data
- Collaboration: Extensive experience working across academia, public health agencies, and industry
- Communication: Scientific writing, technical documentation, and presentations for both technical and non-technical audiences

Education

- **North Carolina State University** — Raleigh, NC
  Ph.D. Candidate in Bioinformatics – Advisors: Dr. Jung-Ying Tzeng & Dr. Wenbin Lu — Expected Aug 2026
  Comajor: Master in Statistics (GPA 3.9)
  Relevant Coursework: Bioinformatics, Statistical Theory/Methods, Neural Networks, Functional Genomics, Computation for Statistical Research, Computational Methods for Molecular Biology
- **Peking University** — Beijing, China
  Master in Epidemiology and Health Statistics, GPA 3.7/4.0 — Aug 2016 – Jul 2018
  Bachelor of Preventive Medicine & Bachelor of Economics, GPA 3.7/4.0 — Aug 2011 – Jul 2016

Experience

- **Research Assistant in Bioinformatics** — Raleigh, NC
  Bioinformatics Research Center, NC State University — Aug 2020 – Present

CNV-profile regression with dual-penalties for DNA copy number variants (Method Development)

- Developed **penalized regression** models using line-graph-guided penalties **(Weighted Fusion) and Lasso** to perform variable selection and smoothness of effect estimation of phenotype-associated CNVs.
- Designed CNV-profile curve function to capture **multiple features (Dosage & Length)** of **DNA structural variants** over a large genomic region and to bypass predefinition of CNV loci.
- Designed simulations to benchmark statistical power and estimation accuracy against existing methods.
- Applied the proposed method to **whole genome sequencing** data from the Alzheimer's Disease Sequencing Project, with **Gene-Ontology annotation** and biological pathway analysis.
  The manuscript under revision at *the Annals of Applied Statistics*

R package development & Scalable Implementation of the above proposed methods

- Released **CNVreg** R package on CRAN implementing genome-wide CNV-profile regression method.
- Achieved **1,600+** downloads by the end of 2025, with full documentation and **reproducible vignette**.

Joint analysis of DNA structural variants and SNPs

- Built scalable computational pipelines to support joint analysis across CNV/SV and SNPs,

- Integrated **haplotype imputation, heritability estimation, rare haplotype clustering, and polygenic risk score** construction to improve association power in whole-genome sequencing studies.

- **Applying Chinese National Health Insurance Claims Data to epidemiological studies.** Beijing, China
  Beijing HealthCom Data Technology Company Jun 2018 – May 2019
  - Performed large data cleaning and feature engineering from **insurance claims** data using SQL on Hadoop.
  - Built reproducible pipelines for epidemiological analysis of **incidence, prevalence**, daily hospital admissions, and direct medical costs by population subgroups.
  - Linked public **climate data** to build time-series models to forecast disease burden: incidence rate, comorbidity, and direct economic burden of diseases.
  - Practiced using **parsing, matching, and Natural Language Processing** to standardize unstructured diagnoses written by doctors and match to International Classification of Diseases 10 codes.
  - Built interactive Tableau dashboards to communicate insights to non-technical stakeholders.
- **CHinese Electronic health Records Research in Yinzhou (CHERRY) study** Zhejiang, China
  Center of Disease Control, Yinzhou District Jun 2015 – Mar 2016
  - Conducted epidemiological **surveillance** for chronic disease management.
  - Integrated **electronic health records, insurance claims, and surveillance data** for longitudinal analysis of incidence, prevalence, and mortality in different subgroups.
  - Built multivariate regression and survival models to identify risk factors and predict lifetime disease risk.

PROJECT

- **Bioinformatics Research Center, North Carolina State University** 2020 – present

Federated Learning for privacy-preserving and effective distributed learning (A review) (Co-Lead)

Performed an in-depth literature review of federated learning frameworks that provide **privacy-preserving** algorithms to learn a valid machine learning model without centralizing the data.

Analyzed the most basic setup of **FedAvg, decentralized FL**, and **robust** aggregation extensions in dealing with practical concerns in FL, including **model poisoning due to attacks and heterogeneity between clients**, with applications to simulated privacy-sensitive data.
Reviewed statistical approaches for heterogeneity-aware inference and efficient communication with copula statistics, surrogate likelihood, and density ratio methods for distributed estimation.

Genome Assembly from PacBio Sequencing Reads (Co-Lead)

Extract potential target reads based on reference genome using DIAMOND blastx and Kraken2;
Performed genome assembly with Hifiasm for 2 target species;
Evaluated assembly **quality** with QUAST and BUSCO, summarizing the genome size, contig N50, and gene completeness and redundancy.

Terrain Identification from wearable device signals using Neural Networks(Co-Lead)

Developed deep learning models (FCNN, CNN, LSTM) using TensorFlow/Keras;
Optimized performance via hyperparameter tuning and validation metrics

Naturally-inspired algorithm for multiple sequence alignment(Co-Lead)

Divided long sequences into more manageable pieces with **Divide & Conquer** using genetic algorithm;
Implemented **Ant Colony Optimization** algorithm for multiple DNA sequence alignment, tuning hyperparameter (such as pheromone and evaporation rate) for optimal alignment paths.

Hands-on experience on Large Language Models and Generative AI (Co-Lead)

Built interactive text and image processing chatbots using ChatGPT and LLaMA APIs.
Developed user-facing web applications with Streamlit and Flask.
Applied fine-tuning and Retrieval-Augmented Generation (RAG) for diagnostic modeling in plant and human disease applications.

- **School of Public Health, Peking University** 2016 – 2018

Disease prevention strategy evaluation based on risk prediction models using EHR and cohort data

- Led a **systematic review and meta-analysis** evaluating effect of statin on cardiovascular disease in China.
- Built sex-specific cardiovascular disease **risk prediction** models using **Cox proportional hazards model** based on an Electronic Health Records database;
  Validated the model in a rural northern Chinese population cohort, assessing predictive capacity via **discrimination and calibration** metrics. (Member)
- Implemented **Markov Chain Monte Carlo simulations** according to the natural history of cardiovascular diseases to evaluate long-term public health intervention strategies with cohort-derived evidence;
  Modeled the effect of **screening and treating** people at high risk of cardiovascular diseases with different strategies: the quantitative strategy based on China-PAR Cardiovascular Diseases Risk Prediction Model and the qualitative strategy recommended by Chinese guideline for prevention of cardiovascular diseases;
  Quantified potential **health and economic impact** of each screening strategy in 10 years: quality-adjusted life years gained and cost-effectiveness ratios to inform decision-making. (Lead)

Publication (see Google Scholar for a full list) and Delivery

- **Si Y.**, Lu W., Tzeng J, et al, CNV-profile regression for CNV association analysis with whole genome sequencing data. **In revision at The Annals of Applied Statistics;**
  Preprint available at bioRxiv (2024) https://www.biorxiv.org/content/10.1101/2024.11.23.624994v1
- **Si Y**, Holloway S, Tzeng, J. CNVreg: An R package for CNV association analysis
  CRAN (2025) https://cran.r-project.org/web/packages/CNVreg/index.html
- Gu X, Guo T, **Si Y.**, et al, Association between ambient air pollution and daily hospital admissions for depression in 75 Chinese cities. American Journal of Psychiatry 177.8 (2020): 735-743. DOI https://doi.org/10.1176/appi.ajp.2020.19070748
- Tian Y., Liu H., Wu Y., **Si Y.**, et al, Association between ambient fine particulate pollution and hospital admissions for cause specific cardiovascular disease: time series study in 184 major Chinese cities. BMJ (2019): 367. DOI https://doi.org/10.1136/bmj.l6572
- Tang X., Zhang D., He L, Wu N, **Si Y.**, etc. Performance of atherosclerotic cardiovascular risk prediction models in a rural Northern Chinese population: Results from the Fangshan Cohort Study. Am Heart J. 2019 May; 211:34-44. https://doi.org/10.1016/j.ahj.2019.01.009
- Zhang D., Tang X., Shen P., **Si Y.**, etc. Multimorbidity of cardiometabolic diseases: prevalence and risk for mortality from one million Chinese adults in a longitudinal cohort study. BMJ Open. (2019) Mar 3; 9(3): e024476. DOI https://doi.org/10.1136/bmjopen-2018-024476
- **Si Y.**, Tang X., Zhang D., etc. [Effectiveness of different screening strategies for primary prevention of cardiovascular diseases in a rural northern Chinese population. Journal of Peking University (Health Sciences), (2018) Jun 18;50(3):443-449] DOI 10.3969/j.issn.1671-167X.2018.03.009

Additional information

- GGB symposium (2022) Best Student Presentation Award
- Teaching assistant: Ph.D-level course ST721 Statistical Genetics (2023)
- Poster (CNV-profile regression for copy number variant association analysis with whole genome sequencing data) presented at the Eastern North American Region (ENAR) Spring Meeting, 2026